Skills Summary

Handling Ambiguous SSA Cases Completely

Use this reference while completing your worksheet or when studying.

SSA means you know A, its opposite side a, and the adjacent side b. The side a swings from the far vertex like a hinge, so it may meet the base line twice, once, or never. Test before you solve, using the height $h = b \sin A$:

- $a < h$: **no** triangle — the side cannot reach.
- $a = h$: **one**, with a right angle at B.
- $h < a < b$: **two** — the ambiguous case.
- $a \geq b$: **one** — B must be acute, since the larger side faces the larger angle.

The law of sines returns only arcsin, which is always acute. The supplement $180° - B$ has the same sine, and it is a second triangle exactly when $A + (180° - B) < 180°$.

1. Test the case first

Compute $h = b \sin A$, then compare h, a and b before touching the law of sines. One multiplication decides how many answers the problem has — and therefore how many you must report. The gap $a - h$ is worth writing down too: it is how much shorter the measured side could have been before the second triangle disappears.

Example: $A = 25°$, $b = 18$, $a = 10$. How many triangles?
Step 1. The count depends only on how the swinging side a compares with the height, so compute h before solving anything.
Step 2. $h = 18 \sin 25°$, and since this is less than $a = 10$, which is in turn less than $b = 18$, the measurement gives two triangles.

$$\boldsymbol{h = 7.61}$$

Try it: $A = 35°$, $b = 20$, $a = 8$. Find h and say how many triangles exist.

check: $\boldsymbol{h = 11.47}$, and $a < h$, so none

2. Two triangles: finish both

Find the acute B_1 from $\sin B = \frac{b \sin A}{a}$, then take $B_2 = 180° - B_1$ and confirm $A + B_2 < 180°$. Each B has its own third angle $C = 180° - A - B$ and its own side $c = \frac{a \sin C}{\sin A}$. Report both triangles.

Example: $A = 32°$, $b = 15$, $a = 10$. Find both values of B.
Step 1. Here $h = 7.95 < a < b$, so two triangles exist and the acute answer alone would be half the solution.
Step 2. $\sin B = \frac{15 \sin 32°}{10} = 0.7949$ gives the acute value below, and its supplement $127.36°$ passes the test since $32° + 127.36° < 180°$.

$$\boldsymbol{B_1 = 52.64°}$$

Try it: $A = 41°$, $b = 22$, $a = 16$. Find the acute B, then its supplement.

check: $\boldsymbol{B_1 = 64.43°}$, so $B_2 = 115.57°$

3. One triangle: show the other candidate fails

When $a \geq b$ the supplement always fails, because $A + (180° - B)$ comes out above 180°. Say so explicitly — "B_2 rejected, $A + B_2 > 180°$" is the line that shows you tested rather than forgot.

Example: $A = 55°$, $b = 9$, $a = 14$. Find B and justify that it is the only one.
Step 1. Since $a = 14$ is longer than $b = 9$, the angle opposite a is the larger one, so B cannot be obtuse and one triangle is guaranteed.
Step 2. $\sin B = \frac{9 \sin 55°}{14} = 0.5266$, and the supplement 148.22° would give $A + B_2 = 203.22° > 180°$, so it is rejected.

$$\boldsymbol{B = 31.78°}$$

Try it: $A = 63°$, $b = 12$, $a = 20$. Find B and say in one line why there is no second triangle.

check: $\boldsymbol{B = 32.32°}$, and $a > b$ rejects the supplement

Four ways this goes wrong.

- Reporting the calculator's arcsin and stopping. It only ever returns the acute angle, so the second triangle never appears unless you look for it.
- Using $a \sin A$ for the height. It is the *fixed* side b that swings down to the base line, so $h = b \sin A$.
- Reaching for the cosine. The height is opposite A in the right triangle that defines it, so it is a sine.
- Rounding B_1 before subtracting it from 180°. Keep full precision until the last step, or B_2, C and c all inherit the error.